

Project Design Phase-II Technology Stack (Architecture & Stack)

Date	01 July 3035
Team ID	23A51A4222
Project Name	Leave Approval System
Maximum Marks	4 Marks

Technical Architecture:

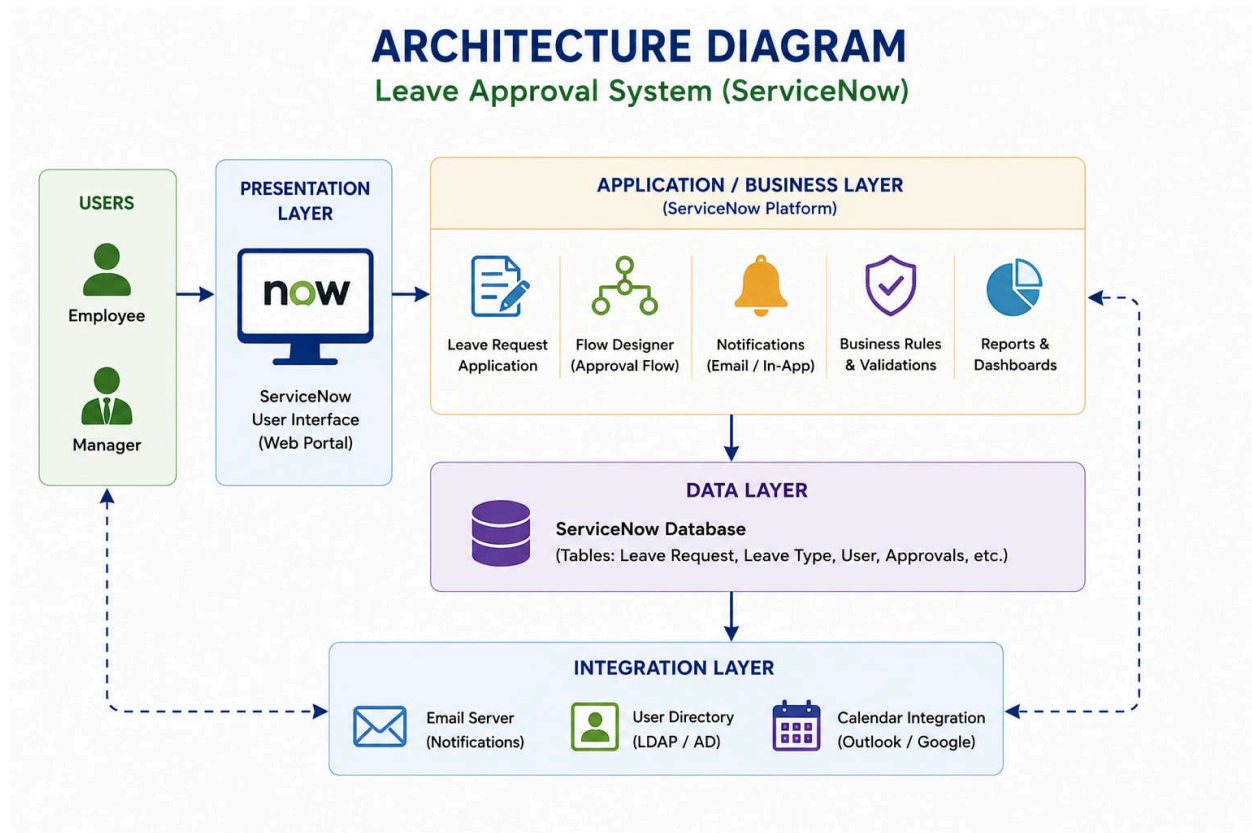


Table-1 : Components & Technologies:

Component	Technology Used	Purpose / Description
Platform	ServiceNow Platform	Provides the foundation for building and running the leave approval system.
Application Development	ServiceNow Studio / App Engine	Used to design and develop custom applications for leave management.
Workflow Automation	Flow Designer / Workflow Engine	Automates the leave request approval process and routing.

User Interface	Service Portal / Forms	Enables employees and managers to interact with the system through intuitive interfaces.
Database	ServiceNow Tables	Stores leave requests, balances, approvals, and history.
Business Logic	Business Rules	Enforces rules such as validation of leave balance and date restrictions.
Client-side Validation	UI Policies / Client Scripts	Ensures data entered by users is valid before submission.
Notifications	Notification Engine / Email / SMS	Sends alerts to employees and managers about leave request status.
Security	Role-based Access Control (RBAC)	Ensures only authorized users can access or act on leave requests.
Reporting & Dashboards	Performance Analytics / Reports	Provides insights into leave trends, balances, and approval statistics.
Integration	IntegrationHub / APIs	Connects with external systems such as HR or payroll for synchronization.

Table-2: Application Characteristics:

Characteristic	Description	How it is Implemented in the System
Purpose	Automates and simplifies the process of requesting and approving employee leave.	Provides a centralized platform for leave submission, approval, notifications, and tracking using ServiceNow.
Users	Employees, Managers and System Administrators are the primary users.	Role-based access using ServiceNow roles (Employee, Manager, Admin).
Accessibility	Accessible anytime, anywhere through web browser.	Built on Service Portal, responsive UI supports access on desktop and mobile devices.
Security	Ensures data security and access control based on user roles.	Implemented using Roles, ACLs, Data Policies and Encrypted connections.
Usability	Easy to use interface for	User-friendly forms, clear

	submitting, approving and tracking leave requests.	navigation, simple dashboards and status indicators.
Reliability	Consistent and accurate processing of leave requests and approvals.	Automated workflows, validations and business rules minimize manual errors.
Performance	Quick response time for smooth user experience.	Optimized workflows, indexed tables and efficient queries ensure fast performance.
Scalability	Supports increasing number of users and leave transactions.	Built on ServiceNow platform which is scalable and cloud-based.
Maintainability	Easy to maintain and update with minimal impact on existing features.	Modular design using best practices, update sets and clear configurations.
Traceability & Reporting	Provides complete tracking and reporting of leave requests and approvals.	Reports and dashboards show leave history, status, approvals and trends.